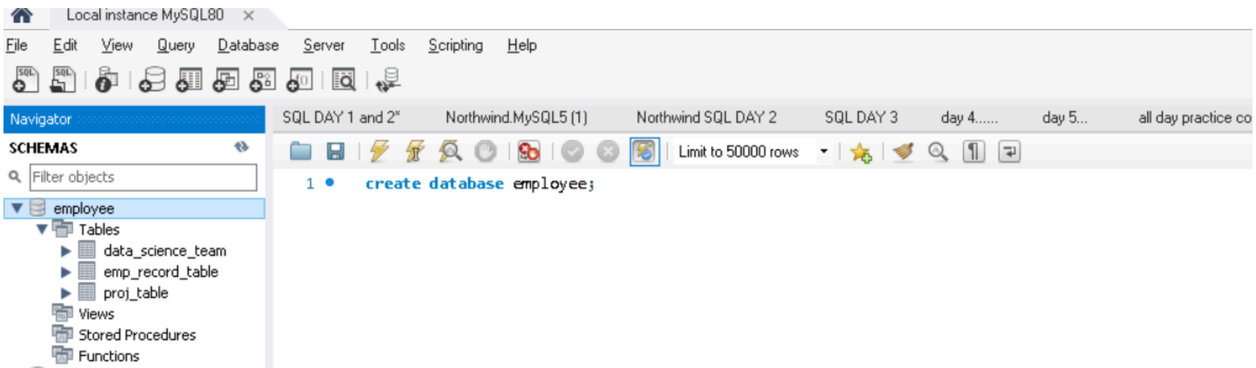
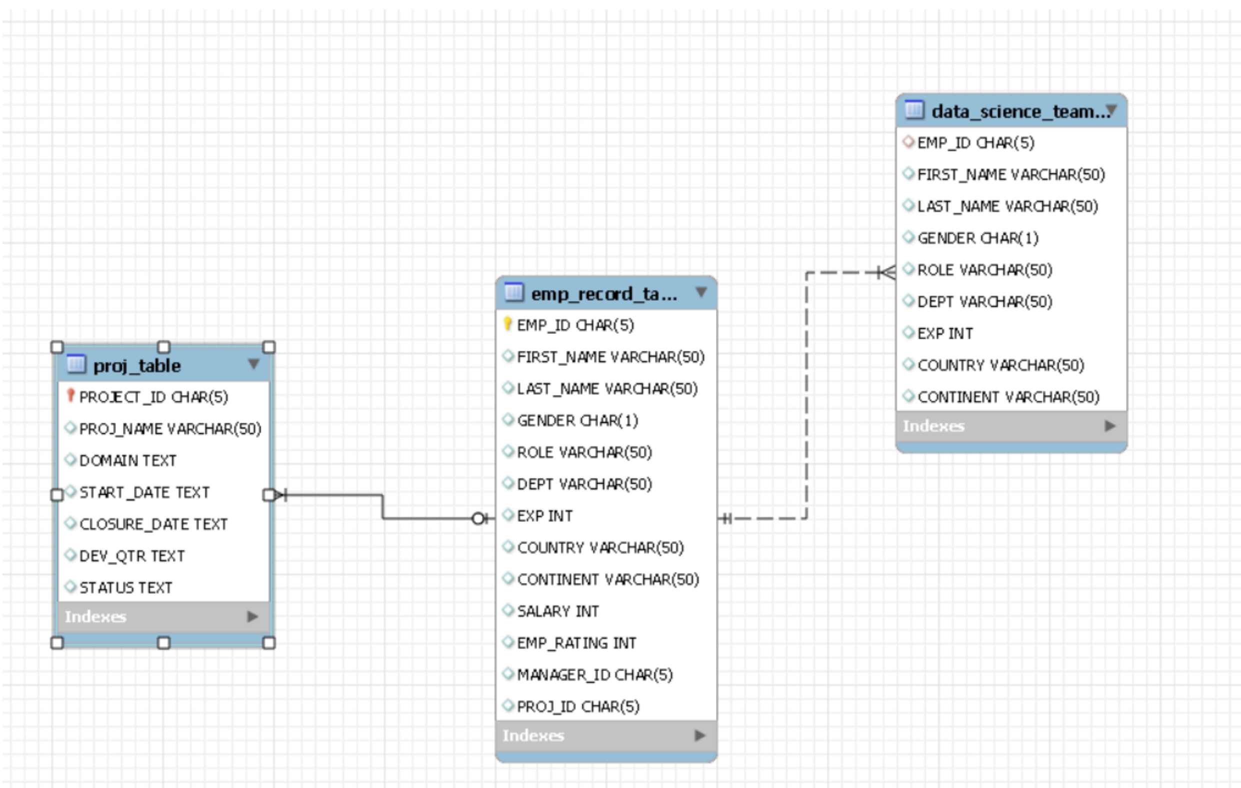
1. Create a database named *employee*, then import **data_science_team.csv proj_table.csv** and **emp_record_table.csv** into the **employee** database from the given resources.



2. Create an ER diagram for the given **employee** database.



3. Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, and DEPARTMENT from the employee record table, and make a list of employees and details of their department.

```sql
-- Task -3
select EMP_ID, FIRST_NAME, LAST_NAME, GENDER,Dept from emp_record_table order by dept;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	Dept
E001	Arthur	Black	M	ALL
E010	William	Butler	M	AUTOMOTIVE
E204	Karene	Nowak	F	AUTOMOTIVE
E428	Pete	Allen	M	AUTOMOTIVE
E532	Claire	Brennan	F	AUTOMOTIVE
E005	Eric	Hoffman	M	FINANCE
E103	Emily	Grove	F	FINANCE
E403	Steve	Hoffman	M	FINANCE
E052	Dianna	Wilson	F	HEALTHCARE
E057	Dorothy	Wilson	F	HEALTHCARE
E083	Patrick	Voltz	M	HEALTHCARE
E505	Chad	Wilson	M	HEALTHCARE
E245	Nian	Zhen	M	RETAIL
E260	Roy	Collins	M	RETAIL
E478	David	Smith	M	RETAIL
E583	Janet	Hale	F	RETAIL
E612	Tracy	Norris	F	RETAIL
E620	Katrina	Allen	F	RETAIL
E640	Jenifer	Jhones	F	RETAIL
NULL	NULL	NULL	NULL	NULL

4 Write a query to fetch EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPARTMENT, and EMP_RATING if the EMP_RATING is:

- less than two
- greater than four

between two and four

```sql
-- Task- 4
select EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT, Emp_Rating,
       case when emp_rating <=2 then 'Low'
       when emp_rating >4 then 'High'
       when emp_rating between 2 and 4 then 'average'
       end as stander
       from emp_record_table;
```

sult Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPT	Emp_Rating	stander
E001	Arthur	Black	M	ALL	5	High
E005	Eric	Hoffman	M	FINANCE	3	average
E010	William	Butler	M	AUTOMOTIVE	2	Low
E052	Dianna	Wilson	F	HEALTHCARE	5	High
E057	Dorothy	Wilson	F	HEALTHCARE	1	Low
E083	Patrick	Voltz	M	HEALTHCARE	5	High
E103	Emily	Grove	F	FINANCE	4	average
E204	Karene	Nowak	F	AUTOMOTIVE	5	High
E245	Nian	Zhen	M	RETAIL	2	Low
E260	Roy	Collins	M	RETAIL	3	average
E403	Steve	Hoffman	M	FINANCE	3	average
E428	Pete	Allen	M	AUTOMOTIVE	4	average
E478	David	Smith	M	RETAIL	4	average
E505	Chad	Wilson	M	HEALTHCARE	2	Low
E532	Claire	Brennan	F	AUTOMOTIVE	1	Low
E583	Janet	Hale	F	RETAIL	2	Low
E612	Tracy	Norris	F	RETAIL	4	average
E620	Katrina	Allen	F	RETAIL	1	Low
E640	Jenifer	Jhones	F	RETAIL	4	average

5.Write a query to concatenate the FIRST_NAME and the LAST_NAME of employees in the *Finance* department from the employee table and then give the resultant column alias as NAME.

```sql
    from emp_record_table;
-- Task -5
select Emp_id, concat(first_name, '',last_name ) Emp_name, Gender, Role, dept from emp_record_table where dept= 'Finance';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Emp_id	Emp_name	Gender	Role	dept
E005	EricHoffman	M	LEAD DATA SCIENTIST	FINANCE
E103	EmilyGrove	F	MANAGER	FINANCE
E403	SteveHoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE

6. Write a query to list only those employees who have someone reporting to them. Also, show the number of reporters (including the President).

```sql
-- Task- 6

select m.First_name,count(*) from emp_record_table e join emp_record_table m on e.manager_id = m.emp_id group by m.First_name;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

First_name	count(*)
Emily	2
Pete	3
Patrick	3
Arthur	5
Janet	3
Tracy	2

7. Write a query to list down all the employees from the healthcare and finance departments using union. Take data from the employee record table.

```sql
-- task -7

select * from emp_record_table where dept = 'healthcare'
union
select * from emp_record_table where dept ='Finance';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA	5000	2	E083	P103
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA	5000	3	E103	P105

8. Write a query to list down employee details such as EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPARTMENT, and EMP_RATING grouped by dept. Also include the respective employee rating along with the max emp rating for the department.

```sql
-- task -8

select emp_id, First_name, Last_name, role, dept, emp_rating,
      max(emp_rating) over(partition by dept) Max_rating from emp_record_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	First_name	Last_name	role	dept	emp_rating	Max_rating
E001	Arthur	Black	PRESIDENT	ALL	5	5
E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	2	5
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5	5
E428	Pete	Allen	MANAGER	AUTOMOTIVE	4	5
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1	5
E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	3	4
E103	Emily	Grove	MANAGER	FINANCE	4	4
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	3	4
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	5	5

9. Write a query to calculate the minimum and the maximum salary of the employees in each role. Take data from the employee record table.

```sql
--
select role, min(salary), max(salary) from emp_record_table group by role;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

role	min(salary)	max(salary)
PRESIDENT	16500	16500
LEAD DATA SCIENTIST	8500	9000
SENIOR DATA SCIENTIST	5500	7700
MANAGER	8500	11000
ASSOCIATE DATA SCIENTIST	4000	5000
JUNIOR DATA SCIENTIST	2800	3000

10.Write a query to assign ranks to each employee based on their experience. Take data from the employee record table.

```sql
-- Task -10

select *, rank() over(order by exp desc) Rank_exp from emp_record_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID	Rank_exp
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL	1
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL	2
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL	3
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL	3
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL	3
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL	6
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204	7
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105	8
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302	9

11. Write a query to create a view that displays employees in various countries whose salary is more than six thousand**.** Take data from the employee record table.

```sql
-- Task-11

CREATE VIEW Vw_Emp_SalaryAbove6k
as
select EMP_ID,FIRST_NAME,LAST_NAME,SALARY,COUNTRY
    from emp_record_table
    where SALARY > 6000
    order by COUNTRY;
select * from Vw_Emp_SalaryAbove6k;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	SALARY	COUNTRY
E103	Emily	Grove	10500	CANADA
E245	Nian	Zhen	6500	CHINA
E583	Janet	Hale	10000	COLOMBIA
E010	William	Butler	9000	FRANCE
E204	Karene	Nowak	7500	GERMANY
E428	Pete	Allen	11000	GERMANY
E260	Roy	Collins	7000	INDIA
E612	Tracy	Norris	8500	INDIA

12. Write a nested query to find employees with experience of more than ten years. Take data from the employee record table.

```sql
-- Task -12

select * from emp_record_table where emp_id in (select emp_id from emp_record_table where exp > 10 );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

emp_record_table 33 ×

13. Write a query to create a stored procedure to retrieve the details of the employees whose experience is more than three years. Take data from the employee record table.

```sql
-- task-13

USE `employee`;
DROP procedure IF EXISTS `emp_exp2`;

DELIMITER $$
USE `employee`$$
CREATE PROCEDURE emp_exp2 ()
BEGIN
select * from emp_record_table where exp > 3;
END$$

DELIMITER ;

call emp_exp2;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5	E428	P204
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2	E583	P109

14. Write a query using stored functions in the project table to check whether the job profile assigned to each employee in the data science team matches the organization's set standard.

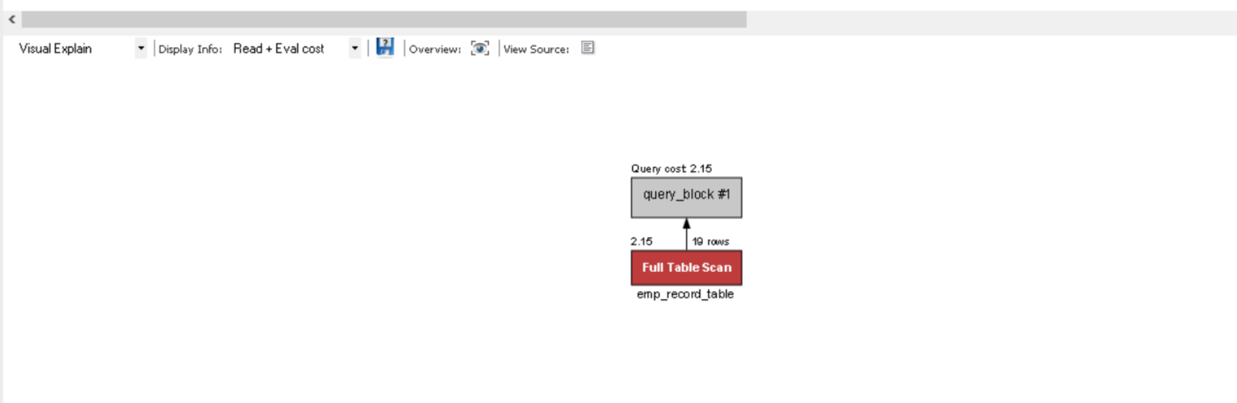
The standard being:
For an employee with experience less than or equal to 2 years assign 'JUNIOR DATA SCIENTIST',
For an employee with the experience of 2 to 5 years assign 'ASSOCIATE DATA SCIENTIST',
For an employee with the experience of 5 to 10 years assign 'SENIOR DATA SCIENTIST',
For an employee with the experience of 10 to 12 years assign 'LEAD DATA SCIENTIST',
For an employee with the experience of 12 to 16 years assign 'MANAGER'.

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	Standard
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	Yes
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	Yes
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	Yes
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	Yes
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	Yes
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	Yes
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	Yes
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA	Yes
E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA	Yes
E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA	Yes
E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE	Yes
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	Yes
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	Yes

15.Create an index to improve the cost and performance of the query to find the employee whose FIRST_NAME is 'Eric' in the employee table after checking the execution plan.

Before Index

```
select * from emp_record_table where FIRST_NAME = 'Eric';
```



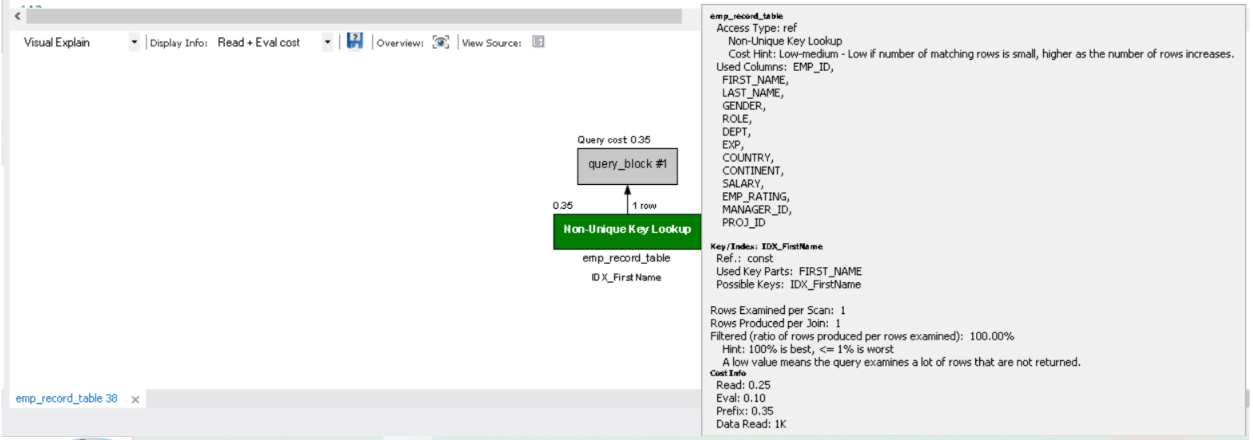
After Index

```sql
-- Task -15

select * from emp_record_table where FIRST_NAME = 'Eric';

CREATE INDEX IDX_FirstName on emp_record_table(FIRST_NAME);

select * from emp_record_table where FIRST_NAME = 'Eric';
```



16. Write a query to calculate the bonus for all the employees, based on their ratings and salaries (Use the formula: 5% of salary * employee rating).

```sql
-- task 16

select EMP_ID,FIRST_NAME,LAST_NAME,SALARY,EMP_RATING,
    (SALARY * .05) * EMP_RATING as Bonus
    from emp_record_table;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

EMP_ID	FIRST_NAME	LAST_NAME	SALARY	EMP_RATING	Bonus
E001	Arthur	Black	16500	5	4125.00
E005	Eric	Hoffman	8500	3	1275.00
E010	William	Butler	9000	2	900.00
E052	Dianna	Wilson	5500	5	1375.00
E057	Dorothy	Wilson	7700	1	385.00
E083	Patrick	Voltz	9500	5	2375.00
E103	Emily	Grove	10500	4	2100.00
E204	Karene	Nowak	7500	5	1875.00
E245	Nian	Zhen	6500	2	650.00
E260	Roy	Collins	7000	3	1050.00
E403	Steve	Hoffman	5000	3	750.00
E428	Pete	Allen	11000	4	2200.00
E478	David	Smith	4000	4	800.00
E505	Chad	Wilson	5000	2	500.00
E532	Claire	Brennan	4300	1	215.00
E583	Janet	Hale	10000	2	1000.00
E612	Tracy	Norris	8500	4	1700.00
E620	Katrina	Allen	3000	1	150.00

17. Write a query to calculate the average salary distribution based on the continent and country. Take data from the employee record table.

```
-- Task -17

select CONTINENT,COUNTRY,Avg(Salary)
    from emp_record_table
    group by CONTINENT,COUNTRY with rollup;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

CONTINENT	COUNTRY	Avg(Salary)
ASIA	CHINA	6500.0000
ASIA	INDIA	6166.6667
ASIA	NULL	6250.0000
EUROPE	FRANCE	9000.0000
EUROPE	GERMANY	7600.0000
EUROPE	NULL	7950.0000
NORTH AMERICA	CANADA	7000.0000
NORTH AMERICA	USA	9440.0000
NORTH AMERICA	NULL	8525.0000
SOUTH AMERICA	COLOMBIA	5600.0000
SOUTH AMERICA	NULL	5600.0000
NULL	NULL	7463.1579